Hotplug DRM pipeline components on non-discoverable video busses

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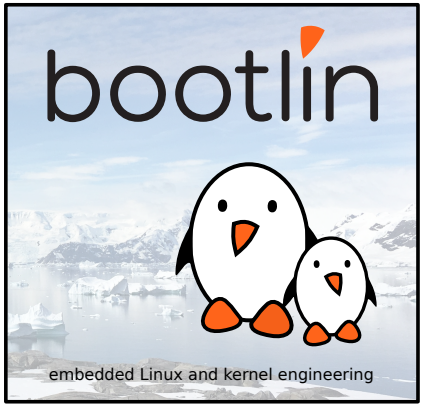

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Corrections, suggestions, contributions and translations are welcome!

Luca Ceresoli

- Embedded Linux engineer at Bootlin
  - Embedded Linux **expertise**
  - **Development**, consulting and training
  - Strong open-source focus
- Linux kernel device driver developer
- Bootloaders, Buildroot and Yocto integration
- Open-source contributor
- Living in **Bergamo**, Italy

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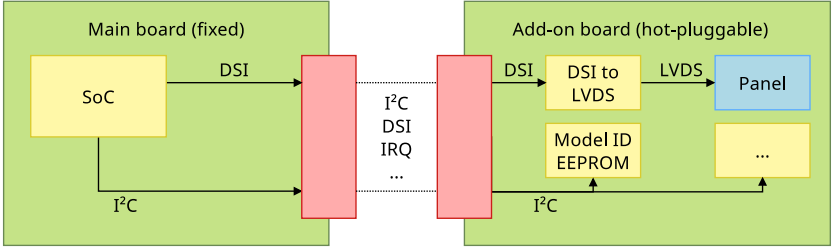
Context

Goal

- A healthcare product, advanced hardware prototype, to be launched in 2025
- Classic ARM64 embedded system, using device tree
  - Works standalone with basic features
- Has a connector for an add-on to extend features
  - Proprietary connector
  - Hot-pluggable by user at any moment
  - Connector uses non-discoverable busses (I2C, MIPI DSI, interrupts)
  - DSI-to-LVDS bridge on the add-on



Overall solution

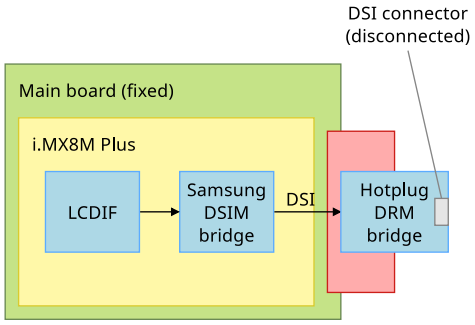
- Device tree + Adding HW = **device tree overlays**
  - Discussion next Friday
  - “Runtime hotplug on non-discoverable busses with device tree overlays”
  - https://lpc.events/event/18/contributions/1696/
- v4: https://lore.kernel.org/all/20240917-hotplug-drm-bridge-v4-0-bc4dfee61be6@bootlin.com/

Proposed approach: the hotplug bridge

The hotplug bridge

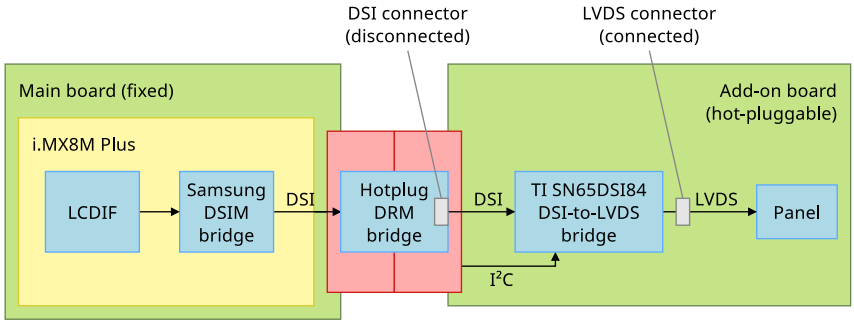
- Main issue with DRM: pipeline is assumed to be immutable (except panel/display after DRM connector)
- Card needed to be always present
- Idea: add a decoupling **“hotplug bridge”**

The hotplug bridge




```
# modetest -c | grep -i '^[a-z0-9]'
Connectors:
id      encoder status          name        size (mm)     modes   encoders
38      0       disconnected    DSI-1       0x0           0       37
```

The hotplug bridge




```
# modetest -c | grep -i '^[a-z0-9]'
Connectors:
id      encoder status          name        size (mm)     modes   encoders
38      0       disconnected    DSI-1       0x0           0       37
39      0       connected       LVDS-1      344x194       1       37
```

The hotplug bridge

- The hotplug bridge is not a chip, it just “represents” the physical connector
- Fully decouples fixed and removable sections
- Existing bridge drivers work unmodified
  - Not aware of hotplug
- Self-contained, only small additions to DRM core (patch 2, patch 3)
- DRM card always available

Current status

- patch 4 adds the `hotplug-drm-bridge` driver
- Ongoing discussion after Sima's review of v2
  - Many changes requested

Work in progress

- Added dynamic LVDS DRM connector [v3]
  - Added/removed on hot(un)plug, like DP MST
- No DRM bridge notifiers [small step in v4]
  - Added `bridge_event_notify()` to `struct drm_bridge_funcs` (patch 2)
- Working on other aspects

Open issues

- Clarifications needed
- What should be moved to the DRM / bridge core?
- Should the hotplug bridge exist (as a `struct drm_bridge`)?
  - If not, how to keep the card existing when there is no add-on?
  - Bridges don't expect next bridge to disappear
- Lifetime, refcounting, ownership issues
  - DRM bridges have no lifetime management or refcounting
  - Let removable bridges be "owned" by the dynamic DRM connector to reuse its ones
  - Need to add refcounting to bridges anyway?
- Bridge removal during atomic updates

Questions? Suggestions? Comments?

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